

GripOS — The AI Operating System for Robotic Manipulation Intelligence

Teaching Machines to Touch the Physical World

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The Problem

We're in the midst of a robotics revolution. Humanoid robots from Tesla, Figure, Apptronik, and 1X are entering factories and warehouses. Industrial arms are everywhere. Delivery bots roam sidewalks. Home robots are finally becoming viable.

But every single one of them faces the same fundamental bottleneck: manipulation.

Getting a robot to reliably pick up, move, and manipulate objects is the hardest unsolved problem in robotics. And it's holding back the entire industry.

The Manipulation Crisis

Why is this so hard?

- **Infinite object variation:** Every object has different shape, size, weight, texture, and fragility
- **Environmental chaos:** Lighting changes, objects move, surfaces vary, clutter accumulates
- **Deformable nightmares:** Fabric, food, cables, packaging — things that change shape
- **Precision requirements:** Sub-millimeter accuracy needed for assembly, surgery, electronics
- **Real-time adaptation:** Must react to slip, unexpected weight, object damage in milliseconds
- **Safety constraints:** Can't crush fragile items or hurt humans

The numbers are brutal:

- 67% of warehouse automation projects fail at the manipulation stage
- Average robot pick rate: 400-800/hour vs. human 1,200+/hour
- 94% of objects robots encounter are "novel" (never seen in training)
- \$12B spent annually on failed automation projects
- 2.1M manufacturing jobs unfilled due to automation limitations

Current solutions don't work:

- **Rule-based systems:** Can't handle variation — break with any new object
- **Computer vision alone:** Can see objects but can't plan grasps reliably
- **End-to-end learning:** Requires millions of examples per object type
- **Simulation-trained models:** Fail dramatically in real-world conditions ("sim-to-real gap")
- **Teleoperation:** Expensive, doesn't scale, defeats the purpose of automation

Every robotics company is building manipulation from scratch. They spend 18-36 months just to get basic picking working. Then they hit a wall when objects change. It's a massive duplication of effort and expertise.

The Solution: GripOS

GripOS is the universal intelligence layer for robotic manipulation.

We provide the AI that turns any robot into a master manipulator — able to grasp, move, and manipulate virtually any object with human-like dexterity.

The Vision

GripOS PLATFORM

PERCEPTION ENGINE

- 3D Object Understanding
- Pose Estimation
- Scene Segmentation
- Material Classification
- Deformation Prediction
- Physics Inference

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GRASP INTELLIGENCE

- Optimal Grasp Generation
- Multi-Finger Coordination
- Failure Recovery
- Force Planning
- Stability Prediction
- Alternative Strategies

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MOTION ORCHESTRATION

- Collision-Free Paths
- Multi-Object Handling
- Assembly Sequences
- Real-Time Adaptation
- Human-Safe Trajectories
- Error Recovery

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TACTILE INTELLIGENCE

- Slip Detection/Prevention
- Texture Recognition
- In-Hand Manipulation
- Force Feedback Control
- Damage Prevention
- Compliant Grasping

CONTINUOUS LEARNING

- Fleet Learning
- Failure Analysis
- Sim-to-Real Bridge
- Demonstration Import
- Cross-Robot Transfer
- Edge-Cloud Sync

HARDWARE ABSTRACTION LAYER
Grippers Arms Hands Humanoids Mobile Platforms

Core Capabilities

- 1. Universal Object Understanding** - Foundation model trained on 500M+ real-world objects - Zero-shot generalization to novel items - Material property inference (fragility, weight, texture) - Deformation prediction for soft objects - Semantic understanding (“handle this like eggs”)
 - 2. Optimal Grasp Generation** - 1000+ grasp candidates generated in <50ms - Ranked by stability, reachability, and task requirements - Works with any gripper/hand morphology - Handles partial occlusion and clutter - Predicts grasp success probability with 96% accuracy
 - 3. Tactile Intelligence** - Real-time force/torque interpretation - Slip detection and prevention (5ms response) - In-hand manipulation without dropping - Damage prevention for fragile items - Works with any tactile sensor hardware
 - 4. Adaptive Motion** - Dynamic replanning as objects/environment change - Collision avoidance with humans and obstacles - Smooth, efficient trajectories - Multi-arm coordination - Assembly sequence optimization
 - 5. Fleet Learning** - Every manipulation attempt across all robots improves the system - Federated learning preserves customer data privacy - Cross-robot knowledge transfer (arm → humanoid → mobile) - Failure cases automatically generate training data - Continuous improvement without downtime
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Product Suite

GripOS Core

The manipulation intelligence platform - Drop-in SDK for any robot platform - ROS2 native with universal adapters - Real-time performance (<20ms latency) - Edge deployment or cloud hybrid - \$25K/robot/year

GripOS Studio

Training and customization environment - Teach new objects via demonstration - Fine-tune for specific materials/tasks - Synthetic data generation - Performance benchmarking - Included with Core

GripOS Sim

High-fidelity simulation environment - Physics-accurate manipulation simulation - Massive parallel training - Automated curriculum generation - Validated sim-to-real transfer - \$10K/seat/year

GripOS Fleet

Enterprise fleet management - Centralized monitoring and analytics - Fleet-wide learning orchestration - Compliance and safety reporting - Integration with WMS/MES - Custom pricing

GripOS Tactile

Hardware-agnostic tactile sensing - Sensor fusion from any hardware - Standardized tactile API - Pre-trained tactile models - Works with existing sensors - Included with Core

Target Markets

Phase 1: Warehouse & Logistics (2026-2027)

\$18B market, growing 25% annually

- E-commerce fulfillment centers
- Third-party logistics (3PL)
- Grocery distribution
- Returns processing

Key customers: Amazon, Walmart, DHL, FedEx, Ocado, XPO

Value proposition: - Handle 95%+ of SKUs (vs. 40% with current solutions) - 3x pick rate improvement - 90% reduction in integration time - Works with existing robot hardware

Phase 2: Manufacturing (2027-2028)

\$32B market for assembly automation

- Electronics assembly
- Automotive components
- Medical device manufacturing
- Consumer goods

Key customers: Foxconn, Tesla, BMW, Medtronic, P&G

Value proposition: - Flexible assembly without reprogramming - Handle multiple product variants - Quality inspection integrated - Rapid line changeover

Phase 3: Food & Agriculture (2028-2029)

\$8B market for food handling automation

- Food processing and packaging
- Fresh produce handling
- Meal kit assembly
- Vertical farming

Key customers: Tyson, Driscoll's, HelloFresh, Plenty

Value proposition: - Gentle handling of delicate items - Sanitary operation - Batch traceability - Variable product accommodation

Phase 4: Humanoid Robots (2028-2030)

\$150B projected market by 2035

- General-purpose humanoids
- Home assistance robots
- Retail/hospitality robots
- Healthcare support robots

Key customers: Tesla, Figure, 1X, Agility, Boston Dynamics

Value proposition: - Turn humanoid hardware into capable workers - Massive acceleration of deployment - Shared learning across humanoid fleet - The Android of robot manipulation

Business Model

Revenue Streams

Stream	Pricing	Year 3 Target
Core Platform	\$25K/robot/year	\$180M
Professional Services	\$300/hour	\$45M
GripOS Sim	\$10K/seat/year	\$25M
Fleet Management	Custom enterprise	\$50M
Training Data Marketplace	Rev share	\$15M
Humanoid OEM Licensing	Per-unit royalty	\$35M
Total		\$350M ARR

Unit Economics

- **CAC:** \$150K (enterprise sales)
- **ACV:** \$500K (20-robot average deployment)
- **Gross Margin:** 82%
- **LTV/CAC:** 8.5x
- **Net Revenue Retention:** 140%
- **Payback Period:** 8 months

The Data Flywheel

Every manipulation attempt generates valuable data:

More Robots → More Data → Better Models → Better Performance → More Robots

We're building the largest manipulation dataset in the world. This creates an insurmountable moat — no competitor can replicate 10 billion real-world grasps.

Competitive Landscape

Direct Competitors

Company	Approach	Weakness
Covariant	AI for warehouse picking	Single use case, proprietary robots only
Osaro	ML-based picking	Limited object generalization
RightHand Robotics	Piece picking systems	Hardware-bundled, narrow focus
Universal Robots	Collaborative arms + software	Basic manipulation, no AI
FANUC	Industrial automation	Rule-based, no learning

Our Advantages

1. **Hardware Agnostic:** Works with any robot, gripper, or sensor — not locked to proprietary hardware
2. **Universal Intelligence:** Handles any object, not trained per-SKU
3. **Tactile-Native:** Built ground-up with force/touch, not vision-only
4. **Foundation Model:** Pre-trained on 500M+ objects, not starting from scratch
5. **Fleet Learning:** Every robot makes every other robot better
6. **Full Stack:** Perception → Planning → Execution → Learning, not point solutions

Strategic Moat

- **Data Network Effects:** Largest manipulation dataset, growing exponentially
- **Cross-Domain Transfer:** Learning from warehouses improves manufacturing
- **Ecosystem Lock-in:** Developers, integrators, and OEMs build on GripOS
- **Talent Concentration:** World's best manipulation researchers
- **Patent Portfolio:** 47 patents on core algorithms

Go-to-Market Strategy

Phase 1: Prove It (Months 1-12)

- Deploy with 3-5 lighthouse customers
- Focus on warehouse picking (clearest ROI)
- Demonstrate 95%+ object handling
- Build case studies and benchmarks
- Target: 50 robots deployed

Phase 2: Scale It (Months 12-24)

- Launch partner program for integrators
- Expand to manufacturing use cases
- Release GripOS Studio for customization
- Aggressive content marketing (benchmark videos, research papers)
- Target: 500 robots, \$25M ARR

Phase 3: Platformize (Months 24-36)

- Launch developer marketplace
- OEM partnerships with humanoid companies
- International expansion (Europe, Asia)
- Acquire tactical sensor company
- Target: 5,000 robots, \$150M ARR

Phase 4: Dominate (Months 36-48)

- Become default manipulation layer for robotics
- IPO or strategic acquisition
- Expand to consumer/home robots
- Launch next-gen foundation model
- Target: 25,000 robots, \$350M ARR

Financial Projections

Metric	Year 1	Year 2	Year 3	Year 4	Year 5
Robots Deployed	50	500	5,000	25,000	100,000
ARR	\$2M	\$25M	\$150M	\$500M	\$1.5B
Gross Margin	70%	78%	82%	85%	87%
Customers	5	35	150	400	1,000
Employees	25	75	200	400	700
Grasps in Dataset	50M	500M	5B	25B	100B

Path to \$1B+ Valuation

- **Year 2:** Series B at \$250M (10x ARR, category leadership)
- **Year 3:** Series C at \$750M (5x ARR, proven scale)
- **Year 4:** Series D at \$2.5B (5x ARR, path to IPO)
- **Year 5:** IPO at \$7.5B+ (5x ARR, market leader)

Team Requirements

Founding Team (5-7 people)

- **CEO:** Robotics industry veteran with enterprise sales experience
- **CTO:** PhD in manipulation/robotics, shipped real robot systems
- **VP Engineering:** Built real-time ML systems at scale
- **VP Research:** Published in top robotics venues (RSS, CoRL, ICRA)
- **VP Sales:** Enterprise automation sales background

Key Hires (Year 1)

- Perception ML Engineers (3-4)

- Motion Planning Engineers (2-3)
- Tactile Sensing Specialists (2)
- Robot Integration Engineers (3-4)
- Enterprise Sales (2)
- Customer Success (2)

Research Partnerships

- Stanford Robotics Lab
- CMU Manipulation Lab
- Berkeley AUTOLAB
- MIT CSAIL

Why Now?

Technology Convergence

1. **Foundation Models:** Transformers enable massive pre-training on manipulation data
2. **Simulation:** GPU-accelerated physics finally enables useful sim-to-real
3. **Tactile Sensors:** Affordable, reliable sensors now available
4. **Compute:** Edge AI chips enable real-time inference on robots
5. **Data:** Robots in production generating unprecedented manipulation data

Market Timing

1. **Labor Crisis:** 2.1M unfilled manufacturing jobs, aging workforce
2. **Humanoid Wave:** Tesla, Figure, 1X shipping robots that need manipulation intelligence
3. **Reshoring:** Companies bringing manufacturing back need automation
4. **E-commerce Growth:** Fulfillment centers desperate for automation
5. **Investor Interest:** \$3.2B invested in robotics in 2025, manipulation is key gap

The Inflection Point

We're at the "iPhone moment" for robotics. The hardware is ready. The AI is capable. The market is desperate. What's missing is the manipulation intelligence layer that makes it all work.

GripOS will be to robots what Android is to phones — the intelligence layer that unlocks the hardware.

Funding & Use of Funds

Seed Round: \$8M

Use	Amount	Purpose
Engineering	\$4M	Core platform development
Research	\$1.5M	Foundation model training
Pilots	\$1M	Lighthouse deployments

Use	Amount	Purpose
Hardware Lab	\$800K	Testing infrastructure
G&A	\$700K	Operations

Series A: \$35M (Month 18)

Use	Amount	Purpose
Engineering	\$15M	Scale platform, build Sim
Sales & Marketing	\$8M	Enterprise GTM
Research	\$6M	Next-gen models
Customer Success	\$4M	Deployment support
G&A	\$2M	Operations

Key Risks & Mitigations

Risk	Mitigation
Sim-to-real gap	Massive real-world data collection from day one
Hardware fragmentation	Hardware abstraction layer, key partnerships
Long enterprise sales	Focus on urgent use cases with fast ROI
Talent competition	Equity, research freedom, impact at scale
Big tech entry	Speed, focus, data moat, integration ecosystem
Safety incidents	Extensive testing, conservative deployment, insurance

The Future: GripOS Everywhere

2030 Vision:

Every robot on Earth uses GripOS for manipulation. From warehouse arms to humanoid workers to home assistants, our intelligence layer is the universal standard.

- 1 million+ robots running GripOS
- 1 trillion manipulation experiences in our dataset
- Robots as dexterous as humans
- \$5B+ in annual revenue
- The “Intel Inside” of robotics

We’re not building a feature. We’re building the foundation of physical AI.

The company that solves manipulation will own the future of robotics.

That company is GripOS.

Contact

For investment inquiries or partnership opportunities:

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“The future is physical. Robots are coming. And they need to learn to touch.”

GripOS — The Hands of the Machine Age